Machine Learning Fundamentals

Synthetic educational document created for RAG development and evaluation practice. It is intended as a controlled development dataset, not as an authoritative source.

1. Introduction to Machine Learning

Machine learning is a branch of artificial intelligence in which computer systems learn patterns from data and use those patterns to make predictions, classifications, or decisions. Instead of writing an explicit rule for every possible situation, a machine learning system learns a model from examples. The quality of a machine learning system depends on the data, features, learning algorithm, and evaluation procedure.

2. Supervised Learning

Supervised learning uses labeled examples. Each training example contains input features and a known target. Regression predicts a numerical value, such as house price or revenue. Classification predicts a category, such as spam or not spam. Common supervised algorithms include linear regression, logistic regression, decision trees, random forests, support vector machines, and gradient boosting.

3. Unsupervised Learning

Unsupervised learning works with data where a target label is not provided. The algorithm attempts to discover structure in the data. Clustering groups similar observations, while dimensionality reduction creates a smaller representation of the original feature space. K-means and principal component analysis are common examples.

4. Training, Validation, and Testing

A dataset is commonly divided into training, validation, and test portions. The training data is used to learn model parameters. Validation data can help select hyperparameters and compare configurations. The test set is kept separate until final evaluation. Using the test set repeatedly during model development can lead to an overly optimistic estimate of performance.

5. Overfitting and Underfitting

Overfitting occurs when a model learns the training data too closely, including patterns that do not generalize to unseen examples. An overfit model may have very strong training performance but weaker test performance. Underfitting occurs when the model is too simple to capture important relationships. Regularization, cross-validation, feature selection, more data, and simpler models can help address overfitting.

6. Evaluation Metrics

For regression, common metrics include mean absolute error, mean squared error, root mean squared error, and R-squared. For classification, common metrics include accuracy, precision, recall, F1-score, and area under the ROC curve. The appropriate metric depends on the problem and the costs of

different types of errors.

7. Feature Engineering

Feature engineering transforms raw data into useful inputs for a model. Examples include scaling numerical variables, encoding categorical variables, extracting dates into useful components, and constructing domain-specific ratios. Feature engineering should be performed carefully to avoid data leakage, where information unavailable at prediction time accidentally enters the training data.